

AhamX Bodhi

Where I Becomes Infinite

Automatic Speech Recognition Models

Module 4.2: Audio and Speech Models

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Session Overview

What You Will Learn

- ASR evolution: HMM-GMM to end-to-end neural
- CTC, seq2seq, and LM-fusion architectures
- Whisper: architecture and capabilities
- State-of-the-art: Whisper v3, Canary, SeamlessM4T
- Engineering ASR pipelines for production

Concept at a Glance

- **Duration:** 9 minutes
- **Bloom's Level:** B2 – Understand
- **Difficulty:** L2 – Intermediate
- **Module:** 4.2 Audio and Speech Models
- **Key metric:** Word Error Rate (WER)





What ASR Must Solve

The ASR Task: Audio → Text

- **Input:** Audio waveform (or mel-spectrogram). Variable length. Contains speech, noise, silence, and overlapping speakers.
- **Output:** Text transcription. Must handle vocabulary of 50,000+ words, proper names, code-switching, and domain jargon.
- **Core challenge:** Acoustic variability – the same word sounds different across speakers, accents, microphones, and environments
- **Key metric:** Word Error Rate (WER) = (Substitutions + Deletions + Insertions) / Total Words. Human WER on clean speech $\approx$ 5–8%.
- **Evaluation benchmarks:** LibriSpeech (audiobooks, clean/noisy), CommonVoice (multilingual), CHiME (noisy environments), AISHELL (Mandarin)

WER below 5% on clean speech is generally considered production-ready for consumer applications. Noisy environments and domain-specific vocabulary require additional engineering.



ASR Architecture Evolution

Three Generations of ASR Models

- **1st generation – HMM-GMM (1970s–2010s):** Acoustic model (Gaussian Mixture Model) + Hidden Markov Model for temporal dynamics + N-gram language model. Kaldi toolkit. High WER, extensive hand-engineering.
- **2nd generation – Deep Learning + CTC (2014–2019):** RNN/CNN acoustic encoder + Connectionist Temporal Classification (CTC) loss. DeepSpeech, ESPnet. First neural end-to-end models. WER 5–10% on clean speech.
- **3rd generation – Transformer End-to-End (2019–present):** Encoder-decoder Transformer with attention-based decoder. Whisper, Conformer, wav2vec 2.0. WER <3% on LibriSpeech clean. Multilingual, multitask.

The shift to Transformer-based end-to-end models eliminated the need for pronunciation dictionaries, language model fusion, and separate acoustic/language model training.



CTC Loss: Enabling End-to-End ASR Training

Connectionist Temporal Classification (Graves et al., 2006)

- **Problem:** Audio frames (100fps) and text tokens are not aligned – we don't know which audio frames correspond to which characters during training
- **CTC solution:** Introduce a "blank" token and sum over all valid alignments between output sequence and target text using dynamic programming
- **CTC output:** Each audio frame produces a probability distribution over vocabulary + blank. Decode by collapsing repeated tokens and removing blanks.
- **Limitation:** CTC is conditionally independent – each output token is predicted independently, ignoring language model context. Combined with external LM for better accuracy.

CTC in Production

CTC-based models (Conformer-CTC, EncDec-CTC) are preferred for streaming ASR – CTC decoders can operate in real-time with fixed latency unlike attention-based seq2seq models.

Whisper: The Standard for Robust ASR



Robust Speech Recognition via Large-Scale Weak Supervision (Radford et al., 2022)

- **Architecture:** Encoder-decoder Transformer. Encoder: 2 Conv layers + ViT-style Transformer on mel-spectrogram. Decoder: autoregressive Transformer generating text tokens.
- **Training data:** 680,000 hours of weakly supervised web audio-transcript pairs in 99 languages – 10× more than prior work. No special data curation.
- **Multitask:** Single model handles ASR, translation (any language → English), language identification, and voice activity detection via special decoder tokens.
- **Model sizes:** Tiny (39M) to Large-v3 (1.5B). Whisper Large-v3 achieves 2.7% WER on LibriSpeech clean.
- **Robustness:** Significantly outperforms commercial ASR on accented speech, technical vocabulary, code-switching – due to diverse training data.

Whisper is the most widely deployed open-source ASR model as of 2025. It is the default choice for production transcription pipelines unless streaming or on-device constraints require smaller models.





State-of-the-Art ASR Models (2024--2025)

Model	WER clean	Languages	Streaming	Notes
Whisper Large-v3	2.7%	99	No	Open, gold standard
Canary-1B (Nvidia)	2.4%	4	No	CTC+Attn hybrid
SeamlessM4T-v2	3.1%	100+	No	Speech-to-speech
Conformer-CTC	1.9%	1	Yes	Streaming-capable
Wav2Vec2-BERT	2.0%	143	No	MMS, low-resource
Deepgram Nova-2	1.7%	1	Yes	Commercial API

Canary-1B (NVIDIA, 2024) uses a multi-task encoder-decoder with CTC auxiliary loss and outperforms Whisper Large-v3 on all 4 supported languages despite having 1/15 parameters

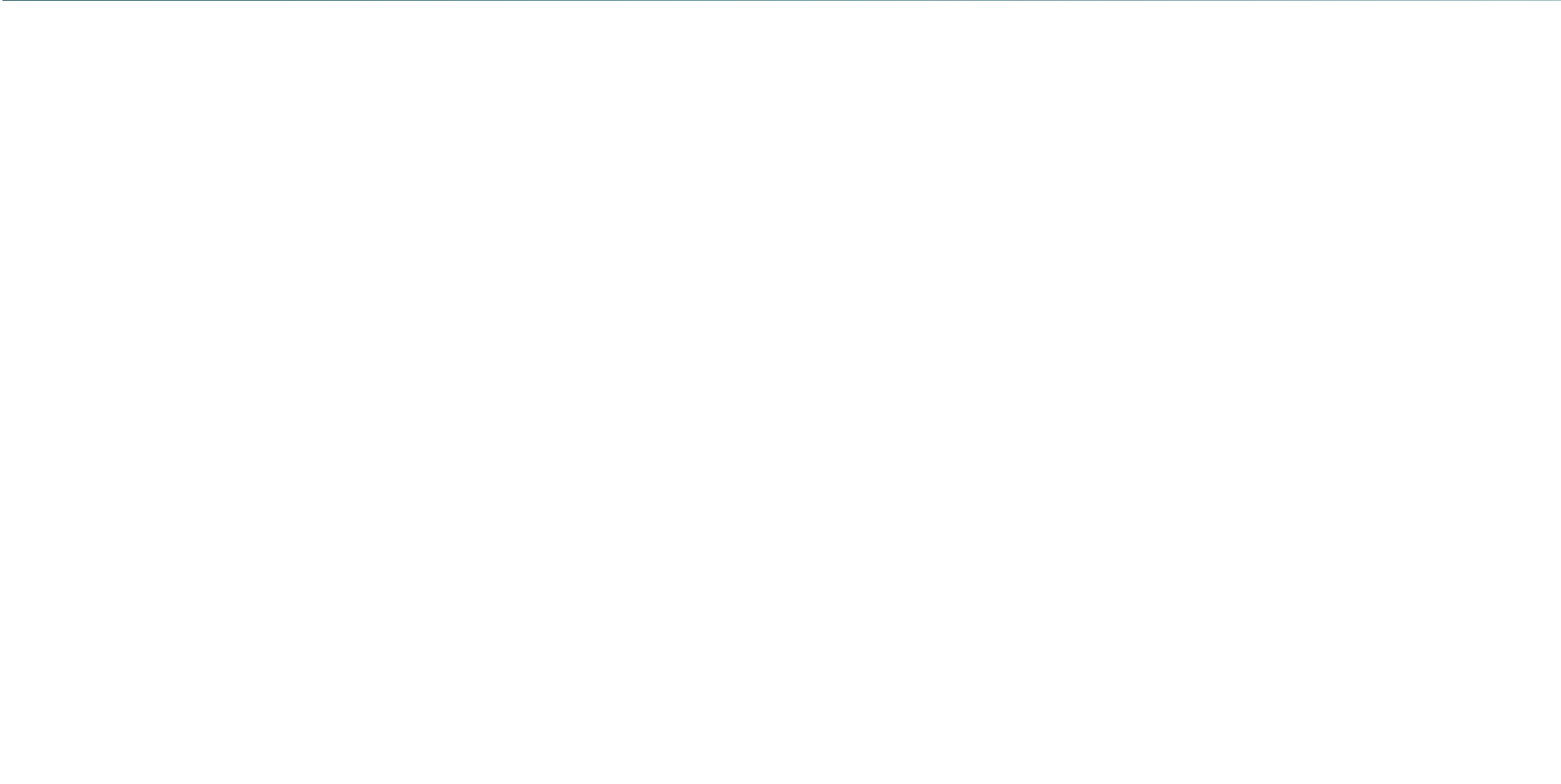


Conformer: CNN + Transformer Hybrid

Conformer (Gulati et al., 2020 – Google)

- **Motivation:** Transformers capture long-range dependencies well but miss local patterns; CNNs excel at local features but struggle with long context – Conformer combines both
- **Architecture:** Each Conformer block: Feed-Forward → Multi-Head Self-Attention → Convolution Module → Feed-Forward (macaron-style)
- **Convolution module:** Depthwise separable convolution captures local acoustic patterns (phoneme-level temporal structure) efficiently
- **Results:** Conformer-Large achieves 1.9% WER on LibriSpeech clean, 3.9% noisy – state-of-the-art at launch, still competitive
- **Streaming variant:** Emformer and Streaming Conformer process audio in chunks with limited right context – enables real-time ASR with controllable latency

The Conformer block is now the standard encoder building block for production ASR – used in Google, Apple, and Amazon ASR systems and in





GenAI Application: Intelligent Meeting Transcription

Building a Production Meeting Intelligence System

- **Pipeline:** Audio recording → VAD (Silero) → Speaker diarization (pyannote-audio 3.0) → Per-speaker ASR (WhisperX) → LLM summarization (GPT-4 / Claude)
- **WhisperX:** Extends Whisper with forced alignment (using wav2vec 2.0) for word-level timestamps – critical for connecting transcription to speaker diarization
- **Output:** Structured JSON: {timestamp, speaker_id, text} for each utterance. LLM post-processes to action items, decisions, and summaries.
- **Commercial products:** Otter.ai, Fireflies.ai, Zoom AI Notes, Microsoft Teams Premium – all use this architecture at scale
- **Latency:** Whisper Large-v3 processes 1 hour of audio in ~2 minutes on A100 GPU. Use Distil-Whisper (6× faster, <1% WER regression) for real-time needs.

Meeting transcription is the highest-volume commercial ASR application – hundreds of millions of meetings per day are transcribed with this pipeline.



GenAI Application: Voice Search and Voice Assistants

Real-Time Voice Query Processing

- **Architecture:** Streaming Conformer-CTC (low latency) → Text → Intent classification or LLM → Response generation → TTS
- **Wake word detection:** Keyword spotting ("Hey Siri", "OK Google") uses tiny ASR models (EfficientNet + CTC, <1MB) running on-device continuously
- **On-device ASR:** Apple uses Whisper-derived models on iPhone for on-device transcription in iOS 17+. Google uses Conformer-based streaming model on Pixel phones.
- **Voice search RAG:** ASR → Text query → Dense retrieval (embedding model) → LLM answer → TTS response. Powers Google Assistant and Amazon Alexa.

On-Device vs. Cloud ASR

- On-device: Distil-Whisper Tiny (39M params, offline, private). Cloud: Deepgram Nova-2 / Google Chirp (lowest WER, highest cost).



GenAI Application: Accessibility and Multilingual ASR

Massively Multilingual Speech (MMS) – Meta AI, 2023

- **Problem:** 7,000+ languages in the world; commercial ASR supports <100. 40% of the world's languages have no digital text resources.
- **MMS approach:** Fine-tune wav2vec 2.0 on 1,107 languages using New Testament religious recordings (uniform domain). Deploy per-language adapter layers (small, efficient).
- **Result:** ASR in 1,107 languages at WER competitive with Whisper on 98 overlapping languages, plus 1,009 additional languages.
- **Impact:** Enables literacy applications, emergency services access, and content creation for speakers of endangered languages
- **AI engineer's role:** Build the serving infrastructure for 1000+ language-specific adapter layers – model routing, lazy loading, and dynamic batching by language

Massively multilingual ASR represents one of the most impactful applications of AI – expanding digital access to communities historically excluded from technology



Key Takeaways

What You Should Remember

- Modern ASR uses end-to-end Transformer encoder-decoders trained with CTC or attention; WER $< 3\%$ on clean speech is routinely achievable
- Whisper (Large-v3) is the open-source standard: 680k hours training, 99 languages, 2.7% WER. Use Distil-Whisper for $6\times$ speedup with minimal accuracy loss.
- Conformer (CNN + Transformer hybrid) is the preferred streaming encoder; CTC loss enables real-time deployment with controllable latency
- Production pipeline: VAD $\rightarrow$ diarization $\rightarrow$ WhisperX $\rightarrow$ LLM post-processing is the current gold standard for meeting transcription
- On-device vs. cloud trade-off: small streaming Conformer for privacy/latency; cloud Whisper Large-v3 or Deepgram Nova-2 for maximum accuracy

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